

Nervous System Interception: A New Paradigm for Haptics

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my approach: **intercepting** nervous system instead of **naively situating** actuators

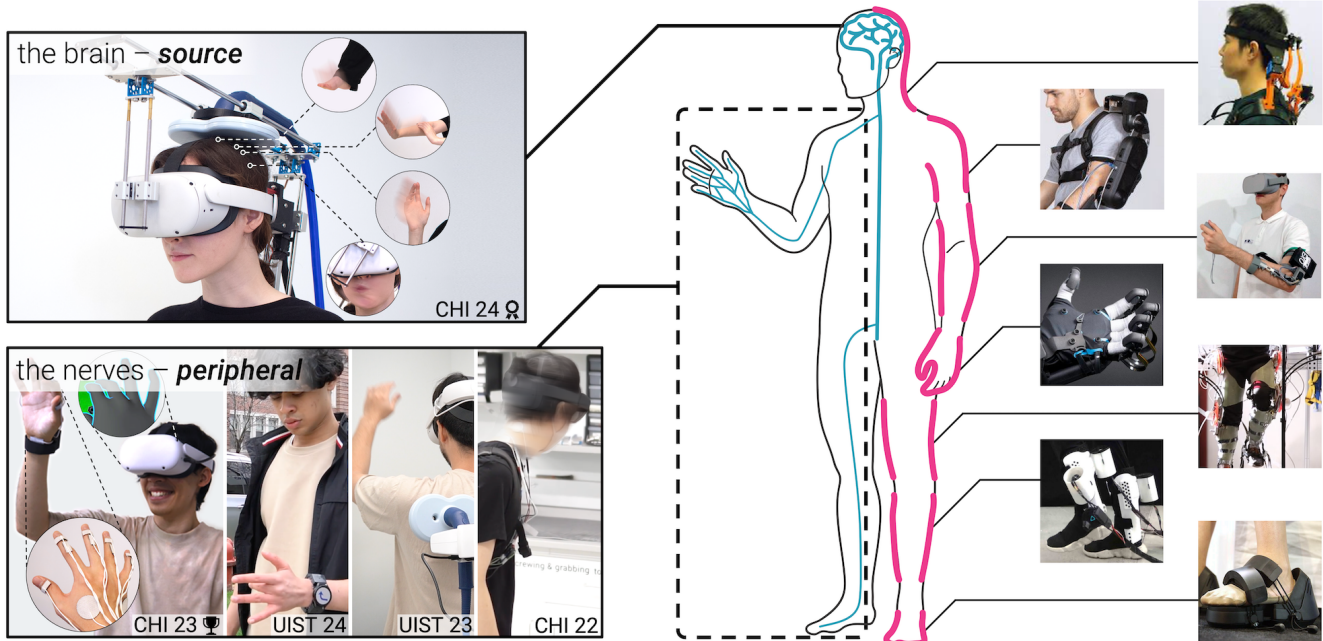


Figure 1: My dissertation work explores interactive devices that intercept the user’s nervous system, positing that these devices pave the way for the next generation of haptic interfaces. Our approach contrasts with most haptic devices today—shown on the right—which are naively situated at the body parts where users should experience feedback (image courtesy: Robotic Neck Brace [21], EduExo Pro [10], Connected Elbow Exoskeleton [4], HaptX [19], Holotron [6], Propelwalker [8], Taclim [3]). In our approach, devices “internally” send electrical signals to the nervous system, creating haptic sensations beyond the point where the device is worn (e.g., stimulating the finger pads from the back of the hand [15]). This allows us to free up the user’s body from devices, blending haptic feedback into real-world interactions, and potentially scaling to a full-body interface by non-invasively stimulating the brain [14].

Abstract

When outputting information to our senses, almost all wearable interfaces follow the same principle: externally generating stimuli (e.g., lights, sounds, vibrations) and then presenting them via devices placed at the endpoints of our sensory system, such as head-mounted displays facing the eyes and vibration motors on the skin. While this intuitive approach of stimulating the endpoints might be sufficient for audiovisual interfaces, we argue that when it comes to engaging the sense of touch (i.e., haptics), it will be insufficient to support a wide variety of interactive experiences.

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Even a single haptic device on the hand will obstruct users from touching or grabbing objects in the real world, making it undesirable for mixed reality. Let alone scaling this approach to a larger portion of the body, which would restrict the user’s whole body. My research introduces an alternative approach to haptic output: instead of stimulating endpoints with external stimuli, we explore interactive devices that “internally” send electrical signals to the user’s nervous system—intercepting the nervous system. Our approach creates haptic sensations beyond the point where the device is worn, establishing a basis for enabling haptic feedback while keeping the user’s body free and scaling haptic interfaces to work for the entire body. This paper provides an overview of our approach: (1) how intercepting the nerves can provide touch and force feedback without obstructing the user’s body with actuators; (2) how it can integrate into practical wearable devices such as a smartwatch; and (3) its potential to eventually generalize to a full-body interface by intercepting the user’s brain.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; • **Hardware** → **Emerging interfaces**.

Keywords

Haptics, Wearable, Nerves, Transcranial Magnetic Stimulation, Electro-tactile, Electrical Muscle Stimulation.

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1 Introduction

Since the advent of HCI, incorporating as many of users' senses into interfaces as possible has been one of the driving forces for research advancement [11]. However, when it comes to engaging the user's sense of touch or bodily movement in interactive experiences, mainstream haptic technology has been stuck with the same principle for thirty years: placing mechanical actuators at the endpoints where users should feel the haptic sensations. This approach is so intuitive and popular that we can find it in haptic devices deployed for various parts of the user's body: haptic gloves for hand sensations [19], upper and lower limb exoskeletons for force feedback [4, 6, 8, 10], head exoskeletons to actuate the head [21], and even devices attached to the sole for the foot haptics [3].

While these might be sufficient for interactions entirely contained in virtual reality simulations, we argue that this approach will be insufficient to cover a wider range of interactive experiences. For example, these devices cover the skin at the attached body location, preventing users from feeling the real world, making them unsuited for mixed reality [2]. Moreover, applying this approach to full-body interactions would tether the user's entire body to numerous actuators [20]. As such, we posit that this is not the way forward to enable the next generation of interactive experiences to go beyond just virtual spaces.

My research introduces an alternative approach to haptic output: instead of stimulating endpoints with external stimuli, **we internally send electrical signals to the nervous system by intercepting the user's nervous system** (Figure 1). In this way, haptic devices can create sensations beyond the point where the device is worn. This approach provides a basis for enabling haptic feedback while keeping the user's body free and scaling haptic interfaces to work for the entire body. In this paper, we present an overview of our approach: (1) how intercepting the nerves can provide touch and force feedback without obstructing the user's body with actuators; (2) how this approach can integrate into practical wearable devices such as a smartwatch; and (3) its potential to generalize to a full-body interface by intercepting the user's brain.

2 INTERCEPTING THE NERVES UNLEASHES THE BODY FROM ACTUATORS (CHI'23 BEST PAPER, CHI'22)

Electrical stimulation applied at one point on a nerve pathway affects receptors downstream [5]—a primary neurological principle we leverage to provide haptic feedback without directly situating

actuators. In our CHI 2023 paper, we demonstrated this concept with a device that provides touch sensations to the front side of the user's hand without attaching any hardware to that location [15]. We achieved this by intercepting the sensory nerves with electrodes placed at the back of the hand (Figure 2). Due to an asymmetry in our tactile sensitivity between the front and back sides, even the stimulation from the back could create ~90% of the sensation in the front. We found that our technique could even create sensations in 11 different locations on the front side while keeping it entirely free for touching and grabbing physical objects. Consequently, this technique enables new applications such as tactile notifications during dexterous activities or VR experiences that rely heavily on physical props.



Figure 2: We achieved tactile feedback on the palmar side of the hand without attaching any actuators to it by intercepting the nerves from the back of the hand. This work received a CHI 2023 Best Paper Award.

While sensory nerves relay information from the skin to the brain, there is the other side of the nervous system where motor neurons and muscles execute commands from the brain to move our body. Looking at the muscles, we can regard them as linkages extending from extremities to joints further up. This is how pairs of electrodes electrically stimulating muscles can apply force feedback to distant extremities—electrical muscle stimulation (EMS), e.g., electrodes attached to the forearm actuate the user's finger movements [1]. Leveraging this principle, in our CHI 2022 paper, we formulated a new way to provide force feedback to the user's head by electrically stimulating the neck muscles [13]. We characterized that our approach could actuate the user's head orientation in the four cardinal directions within a 10-degree error (Figure 3). Having no actuators attached to the user's head, this approach also lowered the bar for combining head-oriented force feedback with AR/VR headsets. We demonstrated this benefit in an AR safety training application where our system guided the user's point of view by directly steering their head orientation, e.g., snapping their view to a fire extinguisher as shown in Figure 3.

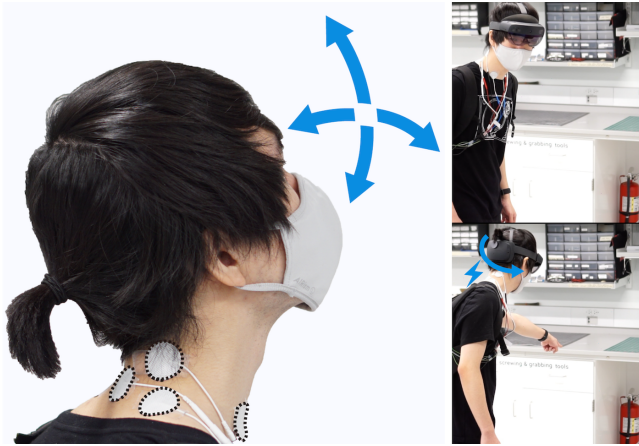


Figure 3: We enabled force feedback to the user’s head without attaching any actuators to it by intercepting the muscles in the neck. This technique can guide the user’s point of view.

3 ENABLING PRACTICAL WEARABLES TO INTERCEPT THE NERVES (UIST’24)

Although intercepting the nerves frees the user’s body from mechanical actuators’ obstruction, it typically requires users to wear electrodes. This additional step poses a barrier to using these devices beyond the laboratory setting.

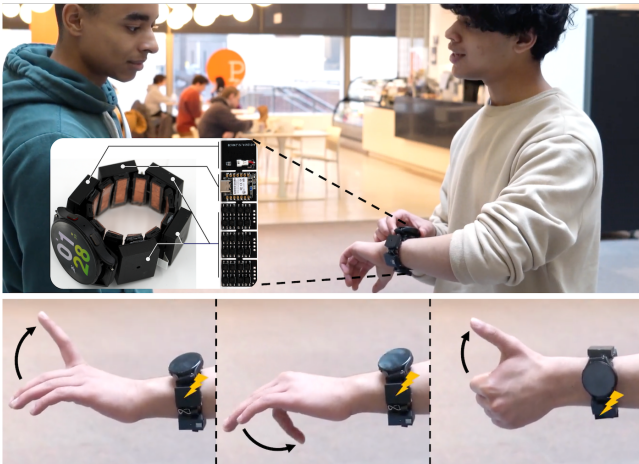


Figure 4: We integrated EMS into a smartwatch, a device that most users already wear on the wrist. This enhanced the practicality & social acceptability of intercepting the nerves.

Therefore, we aimed to enhance the practicality of this class of haptic devices by integrating them into a familiar wrist-worn form factor [12, 18]. Particularly, in our latest UIST paper [12], we engineered an EMS device that is conveniently integrated into a smartwatch (Figure 4). We discovered that, while forearm stimulation is the powerhouse for finger muscles, stimulating the wrist can still reliably contract some of these muscles. We validated that our device could actuate the thumb, index, middle, and pinky fingers, as well as flexing the wrist. Moreover, we found that users felt more comfortable wearing our device in social settings with this more

familiar and practical form factor. We believe that this approach opens up new smartwatch-based interactions, such as enabling a smartwatch to guide the user’s body and output force feedback during everyday tasks (e.g., playing music or exercising).

4 INTERCEPTING THE NERVES WITHUOT ELECTRODES (UIST’23)

The works covered so far used electrical stimuli via electrodes attached to the skin in common, but electrodes are not the only method to intercept the nerves. Looking at how neuroscientists study our nervous system, it is possible to excite the nerves without attaching electrodes by intersecting an oscillating magnetic field created by a coil with the user’s body—magnetic stimulation. In our UIST 2023 paper, we explored how interactive systems can benefit from switching from electrical to magnetic muscle stimulation (MMS) [16]. Through a series of user studies, we discovered that: (1) MMS could apply force feedback to muscles even when the coil is up to 5 cm away from the body; (2) the skin tingling sensation (a side effect of nerve stimulation) from MMS was 50% less than that from EMS; (3) force feedback via MMS was perceived as more realistic and clearer compared to EMS. We demonstrated that, with these benefits, users were able to experience force feedback without wearing any devices, e.g., by just sitting on a chair or having a robotic arm move the MMS coil (Figure 5).

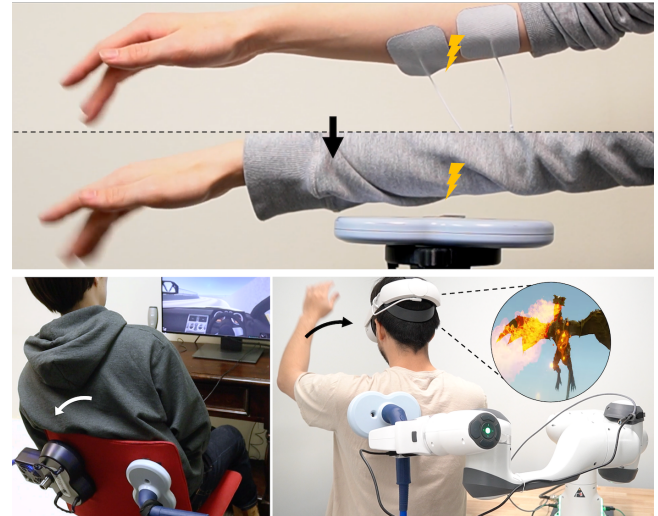


Figure 5: We explored how interactive systems can benefit from switching electrical to magnetic muscle stimulation.

5 INTERCEPTING THE BRAIN SCALES TO A FULL-BODY HAPTIC INTERFACE (CHI’24 HONORABLE MENTION)

Having developed interactive systems that intercept nerves mid-pathway, we naturally began to wonder **how an interactive system would look like if we moved beyond the midpoint**. The logical next step was to actuate the source of the nervous system (i.e., the user’s brain) to generate sensations across the entire body. In our CHI 2024 paper, we presented this vision, which we termed

“haptic source-effector” [14]. Haptic source-effector provides touch and force feedback to multiple parts of the user’s body by stimulating the user’s brain cortex. We implemented this using transcranial magnetic stimulation (TMS)—a non-invasive technique from neuroscience and medicine where electromagnetic pulses safely stimulate brain areas. We characterized that this approach could render ~15 touch and force feedback sensations throughout the body (e.g., hands, arms, legs, feet, and jaw) by stimulating the user’s sensorimotor cortex with a single magnetic coil moved mechanically across the scalp (Figure 6).

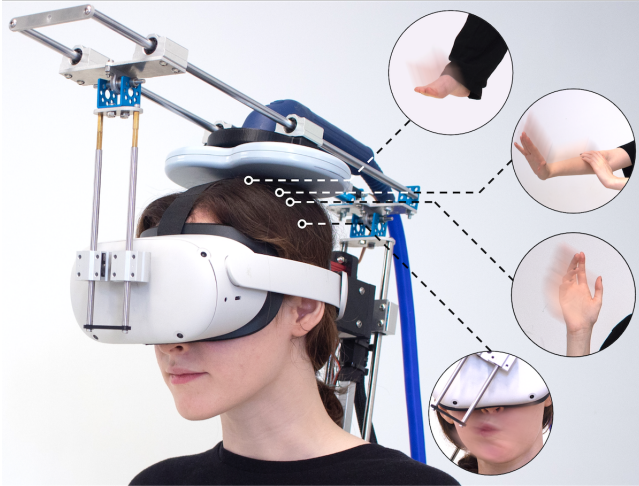


Figure 6: We envisioned “haptic source-effector”, which creates haptic sensations across the user’s entire body. This represents the ultimate vision of intercepting the nervous system. This work received a CHI 2024 Best Paper Honorable Mention.

Haptic source-effector clearly illustrates the contrast between intercepting the nervous system and traditional haptic approaches that individually attach actuators to the endpoints. While the proposed head-worn device is currently bulky, we believe that the proposed vision of centralizing haptic feedback across the full body into a single device over the brain paves the way for the next generation of haptics.

6 INSIGHTS

Intercepting the nervous system vs. attaching to the endpoints. Whether it be touch or force feedback, intercepting the nervous system provides feedback to body locations where the user does not wear the device. This benefit particularly shines in interactive situations where users interact with both virtual and physical interfaces (mixed reality) or in haptic experiences situated in users’ real-world activities. On the contrary, traditional haptic end-effectors have advantages in feedback’s spatial resolution and the variety of feedback modalities (e.g., skin-stretching, thermal feedback). As such, end-effectors are more suited when the goal of the interface is high-fidelity haptic simulation.

Intercepting the source vs. the periphery. While the brain is the source of the nervous system, it does not mean that the haptic source-effector rules out other methods of intercepting nerves at

the periphery. For instance, when it comes to the spatial resolution of haptic feedback, as we move further from the endpoint, the nerves become more bundled, making it more difficult to create sensations in smaller regions. As we observed, intercepting nerves at the back of the hand could create sensations in more than 10 locations on the front side of the hand [15], whereas the haptic source-effector techniques only stimulated a single point on the hand [14]. Although technology is likely to advance for both peripheral and source approaches, this trade-off should guide the choice of an appropriate approach depending on the context.

Trade-off in stimulation modalities. Magnetic muscle stimulation and haptic-source effector demonstrated a coil creating a magnetic field as an alternative to using electrodes. While magnetic stimulation improves the quality of haptic sensation and enables stimulation without directly attaching a device, a stimulation coil is significantly bulkier than electrodes. As such, for most wearable & mobile haptic applications today, it is more practical to use electrical stimulation via electrodes. However, magnetic stimulation offers a promising avenue to explore novel principles in intercepting the nervous system that are not feasible with electrodes.

7 FUTURE WORK

Extending the novel techniques we have developed to intercept the nervous system, we are currently exploring how our approach, especially when applied to the brain, could assist users with their own body movements and potentially facilitate learning physical skills. Moreover, using our interactive systems as a testbed, we are also collaborating with neuroscientists to probe into the user’s cognitive states, such as the sense of agency, while experiencing feedback from our devices.

8 RESEARCH AND DISSERTATION STATUS

I have just completed my fourth year as a PhD student in the University of Chicago’s Department of Computer Science under the advisement of Professor Pedro Lopes. I have published the aforementioned six papers on my dissertation topic (five as first author, one as second author), and two papers on haptics more broadly [7, 9]. Alongside my publications, I spend time fostering a community for researchers at the intersection of neuroscience and HCI, which led to co-organizing the “NeuroCHI” panel at CHI 2024 [17]. Receiving critique and engaging in discussions with the doctoral symposium panel would greatly assist me in learning how to effectively present my work.

Acknowledgments

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References

- [1] Xueliang Bao, Yuxuan Zhou, Yunlong Wang, Jianjun Zhang, Xiaoying Lü, and Zhigong Wang. 2018. Electrode placement on the forearm for selective stimulation of finger extension/flexion. *PLoS one* 13, 1 (2018), e0190936.
- [2] Carlos Bermejo and Pan Hui. 2021. A survey on haptic technologies for mobile augmented reality. *ACM Computing Surveys (CSUR)* 54, 9 (2021), 1–35.
- [3] Cerevo. [n. d.]. Taclim - Cerevo. <https://taclim.cerevo.com/ja/>
- [4] Daniel H de la Iglesia, André Sales Mendes, Gabriel Villarrubia González, Diego M Jiménez-Bravo, and Juan F de Paz Santana. 2020. Connected elbow exoskeleton system for rehabilitation training based on virtual reality and context-aware. *Sensors* 20, 3 (2020), 858.
- [5] Johanna C Forst, Derek C Blok, Julia P Slopesma, John M Boss, Lane A Heyboer, Carson M Tobias, and Katharine H Polasek. 2015. Surface electrical stimulation to evoke referred sensation. *Journal of Rehabilitation Research & Development* 52, 4 (2015).
- [6] Holotron. [n. d.]. Holotron. <https://www.holotron.com/>
- [7] Arata Jingu, Yudai Tanaka, and Pedro Lopes. 2023. Lipio: Enabling lips as both input and output surface. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [8] Pingchuan Ke, Shaoyu Cai, Haichen Gao, and Kening Zhu. 2022. Propelwalker: A leg-based wearable system with propeller-based force feedback for walking in fluids in vr. *IEEE Transactions on Visualization and Computer Graphics* 29, 12 (2022), 5149–5164.
- [9] Jun Nishida, Yudai Tanaka, Romain Nith, and Pedro Lopes. 2022. DigituSync: A dual-user passive exoskeleton glove that adaptively shares hand gestures. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*. 1–12.
- [10] Eduexo Pro. [n. d.]. Exoskeleton | EduExo Pro. <https://www.auxivo.com/eduexo-pro>
- [11] Ivan E Sutherland et al. 1965. The ultimate display. In *Proceedings of the IFIP Congress*, Vol. 2. New York, 506–508.
- [12] Akifumi Takahashi, Yudai Tanaka, Tamhane Archit, Alan Shen, Shan-Yuan Teng, and Pedro Lopes. 2024. Can a Smartwatch Move Your Fingers? Compact and Practical Electrical Muscle Stimulation in a Smartwatch. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–15.
- [13] Yudai Tanaka, Jun Nishida, and Pedro Lopes. 2022. Electrical head actuation: Enabling interactive systems to directly manipulate head orientation. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [14] Yudai Tanaka, Jacob Serfaty, and Pedro Lopes. 2024. Haptic Source-Effector: Full-Body Haptics via Non-Invasive Brain Stimulation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–15.
- [15] Yudai Tanaka, Alan Shen, Andy Kong, and Pedro Lopes. 2023. Full-hand electro-tactile feedback without obstructing palmar side of hand. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [16] Yudai Tanaka, Akifumi Takahashi, and Pedro Lopes. 2023. Interactive Benefits from Switching Electrical to Magnetic Muscle Stimulation. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–12.
- [17] Yudai Tanaka, Angela Vujic, Pattie Maes, Robert JK Jacob, Olaf Blanke, Sho Nakagome, and Pedro Lopes. 2024. NeuroCHI: Are We Prepared for the Integration of the Brain with Computing?. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–5.
- [18] Yudai Tanaka, Neil Weiss, Robert Cole Bolger-Cruz, Jess Hartcher-O'Brien, Brendan Flynn, Roger Boldu, and Nicholas Colonnese. 2024. ReaWristic: Remote Touch Sensation to Fingers from a Wristband via Visually Augmented Electro-Tactile Feedback. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE.
- [19] Scott Varga. [n. d.]. HaptX: Haptic gloves for virtual reality and robotics. <https://haptx.com/>
- [20] Tae-Heon Yang, Jin Ryong Kim, Hanbit Jin, Hyunjae Gil, Jeong-Hoi Koo, and Hye Jin Kim. 2021. Recent advances and opportunities of active materials for haptic technologies in virtual and augmented reality. *Advanced Functional Materials* 31, 39 (2021), 2008831.
- [21] Haohan Zhang and Sunil K Agrawal. 2017. An active neck brace controlled by a joystick to assist head motion. *IEEE Robotics and Automation Letters* 3, 1 (2017), 37–43.